

RMCNET: REFLECTION AND MOIRÉ REMOVAL FOR VIRTUAL PRODUCTION

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ABSTRACT

Virtual production with large-scale digital screens has revolutionized filmmaking. However, the smooth glassy surfaces of the screens, coupled with complex camera–display interactions, introduce challenging degradations, *i.e.*, screen reflections and moiré patterns, which severely compromise visual quality. Existing approaches for demoiréing and reflection removal fall short in this context, as they treat these degradations in isolation. To address this challenge, we present the Screen Degradation Dataset (SDD), the real-world dataset that jointly captures both degradations. We further propose the Reflection-Moiré Complement Network (RMCNet), which integrates a Reflection Separation Module (RSM) to disentangle reflection and transmission layers and a Reflection-Moiré Complement Module (RMCM) to exploit complementary cues between the two degradation types. Extensive experiments demonstrate that RMCNet surpasses state-of-the-art methods in both quantitative and qualitative evaluations.

Index Terms— Virtual Production, Screen Reflection Removal, Moiré Pattern Removal, Image Restoration

1. INTRODUCTION

In the contemporary landscape of digital imagery and virtual environments, virtual production has emerged as a transformative technique in visual media production. By using large-scale screens to simulate backgrounds, it offers a more flexible, economical, and efficient alternative to traditional green-screen or on-location methods. Virtual production integrated to modern film pipelines, enables real-time visualization, interactive lighting, and immersive set design, giving filmmakers unprecedented creative control [1]. However, this approach also introduces technical challenges that can compromise image quality. When cameras capture content displayed on digital screens, visual degradations often emerge. One of the degradation is moiré patterns [2], arising from the interaction between the camera’s sensor frequency and that of the digital screen, leading to the appearance of undesirable fringes in the captured images. Additionally, the smooth glass composition of screens poses a problem by reflecting ambient

light directly into the camera lens, further degrading the quality of the imagery and obscuring the content being captured.

The advancement of deep learning has led to significant improvements in image demoiréing techniques [2, 3]. Nonetheless, these methods overlook the prevalent issue of specular reflections on digital screens. While Single-Image Reflection Removal (SIRR) methods [4, 5] have shown success in separating reflection and transmission layers, they typically require ground-truth supervision for both layers, an impractical demand in real-world scenarios, and are not tailored for screen reflections. More broadly, deep learning-based image restoration has achieved impressive results across diverse degradations such as blur, noise, and raindrops [6–11]. Yet, effectively addressing digital screen moiré patterns and reflections remains challenge.

In this study, we introduce the Real-World Screen Degradation Dataset (SDD), designed to enable exhaustive research into the phenomena of digital screen reflections and moiré patterns, alongside their remediation. Unlike prior efforts that often treat these degradations in isolation or rely on synthetic data, the SDD captures both phenomena under real-world conditions. Benchmarking results in Table 2 demonstrate that the combination of demoiréing and SIRR, together with other state-of-the-art (SOTA) restoration methods, remains markedly inadequate, thereby underscoring the inherent difficulty of this task. To address this gap, we propose the Reflection Moiré Complementary Network (RMCNet). RMCNet begins with a Reflection Separation Module (RSM) that employs a shared encoder and dual decoders to disentangle the input into reflection and transmission layers. At its core lies the Reflection Moiré Complementation Module (RMCM), which integrates complementary cues across spatial and channel dimensions in the bottleneck stage, enabling more accurate reconstruction by leveraging the synergy between reflection and moiré features. Furthermore, we conduct controlled experiments on synthetic data to illustrate the complexity of real-world degradation and to validate the unique contribution of the SDD. Experimental results demonstrate that our approach achieves substantial improvements in image quality for screen-captured content.

Our main contributions can be summarized as follows: (1) We develop real world dataset aimed at facilitating research into digital screen reflections and moiré pattern degra-

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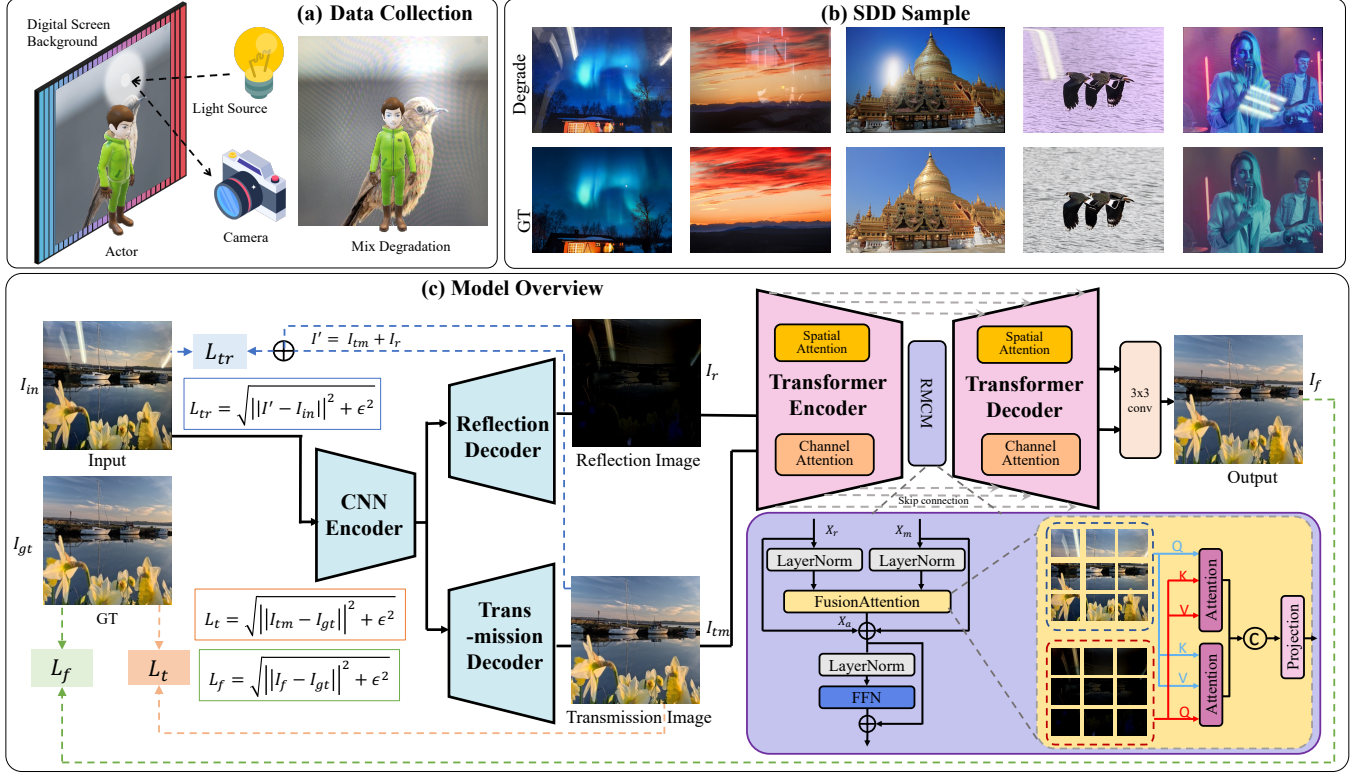


Fig. 1. Overview of the proposed framework and dataset. (a) Data collection: screen background with actor and lighting introduces reflection and moiré. (b) SDD examples: degraded (top) and ground truth (bottom). (c) Model architecture.

dation. (2) We propose Reflection-Moiré Complement Network (RMCNet) to effectively eliminate both screen reflection and moiré patterns with the Reflection Separation Module (RSM) and introduce the Reflection Moiré Complementation Module (RMCM) to enhance the overall restoration. (3) We conduct experiments and compare our scheme with existing SOTA methods, and the results demonstrate the superiority of our proposed approach.

2. PROPOSED METHOD

2.1. Screen Degradation Dataset (SDD)

To address the challenges of digital screen degradation, particularly reflections and moiré patterns, and to facilitate both model training and method evaluation, we introduce the Screen Degradation Dataset (SDD). The dataset consists of 1,990 real-world image pairs captured using smartphones and cameras mounted on tripods, targeting LED screens commonly employed in filmmaking, as illustrated in Figure 1(a). During collection, a variety of images were displayed on the screens while artificial light sources were positioned and adjusted to simulate different real-world conditions. After each capture, lighting intensity, color, and position were varied to increase diversity and ensure comprehensive coverage of reflection and glare scenarios. To enrich the dataset and expand its potential utility, we further synthesized 4,000 reflection-

moiré images using a public demoiré dataset [2], forming a subset named S-SDD. This addition magnifies the overall dataset size and provides broader training possibilities, while also demonstrating that real-world degradations remain more challenging to address. The SDD is divided into 1,730 training pairs and 260 testing pairs, carefully selected to represent a broad range of degradation cases, and to ensure precise alignment of paired images despite variations in lighting and screen properties, the preprocessing pipeline incorporates the Scale-Invariant Feature Transform (SIFT) algorithm [12].

2.2. Reflection-Moiré Complement Network(RMCNet)

The overall pipeline is illustrated in Figure 1 (c). The input is a screen-degraded image $I_{in} \in \mathbb{R}^{3 \times H \times W}$ containing both moiré patterns and reflections. We first employ the Reflection Separation Module (RSM), which consists of a shared encoder followed by two decoders. This module decomposes the input into a reflection layer I_r and a transmission layer I_{tm} . Subsequently, both I_r and I_{tm} are processed in parallel by their respective transformer branches. Each branch incorporates three channel-attention transformer blocks and one spatial-attention transformer block in the encoder, followed by channel-wise concatenation. The concatenated features are then passed through a merge transformer layer and projected via a convolutional layer into moiré features X_{tm} .

and reflection features $\mathbf{X}_r$. These feature representations are further refined in the Reflection-Moiré Complementary Module (RMCM), located at the bottleneck, to facilitate complementary information exchange between the two degradation types. Afterward, the two feature streams are fed into their corresponding decoders for reconstruction. Finally, the reconstructed paths are concatenated and projected through a 3×3 convolutional layer to generate the restored output. To enhance recovery, skip connections are employed by concatenating encoder features with their corresponding decoder features.

Reflection Separation Module (RSM). The RSM is designed to disentangle reflection and transmission components from a reflection-degraded input image. Structurally, the RSM adopts a U-Net like architecture [13], consisting of a shared encoder followed by two task-specific decoders. One decoder reconstructs the reflection layer $\mathbf{I}_r$, while the other generates the transmission layer $\mathbf{I}_{tm}$, corresponding to the reflection-free content. To guide the learning process, we introduce two complementary losses. First, the transmission loss L_t enforces fidelity between the predicted transmission $\mathbf{I}_{tm}$ and the ground truth image $\mathbf{I}_{gt}$. Second, the reconstruction loss L_{tr} ensures that the input image $\mathbf{I}_{in}$ can be approximately recovered by combining the predicted reflection and transmission layers through element-wise addition, *i.e.*, $\mathbf{I}' = \mathbf{I}_r + \mathbf{I}_{tm}$. Formally, the loss functions are defined:

$$\begin{aligned} L_t(\mathbf{I}_{tm}, \mathbf{I}_{gt}) &= \sqrt{\|\mathbf{I}_{tm} - \mathbf{I}_{gt}\|^2 + \epsilon^2}, \\ L_{tr}(\mathbf{I}', \mathbf{I}_{in}) &= \sqrt{\|\mathbf{I}' - \mathbf{I}_{in}\|^2 + \epsilon^2} \end{aligned} \quad (1)$$

where ϵ is a small constant added for numerical stability. Together, these objectives encourage the RSM to produce reflection and transmission layers that are not only structurally disentangled but also consistent with the input image.

Reflection-Moiré Complement Module (RMCM). We design a transformer-based Reflection-Moiré Complement Module (RMCM) to jointly leverage the complementary information from reflection and moiré features while alleviating their individual shortcomings. Let $\mathbf{X}_m$ denote the moiré feature representation and $\mathbf{X}_r$ denote the reflection feature representation. First, we employ a cross-fusion attention mechanism to adaptively integrate moiré and reflection cues, formulated as:

$$\begin{aligned} \mathbf{X}_a &= \text{FusionAttention}(\text{LN}(\mathbf{X}_h), \text{LN}(\mathbf{X}_m)), \\ \mathbf{X} &= \text{LN}(\mathbf{X}_r) + \text{LN}(\mathbf{X}_m) + \mathbf{X}_a, \\ \tilde{\mathbf{X}} &= \text{GELU}(\text{FFN}(\text{LN}(\mathbf{X}))) + \mathbf{X}, \end{aligned} \quad (2)$$

where $\text{LN}(\cdot)$ denotes layer normalization, $\text{FFN}(\cdot)$ is a feed-forward network, and $\text{GELU}(\cdot)$ is the GELU activation [14] function. The residual connection ensures stable gradient propagation and facilitates the preservation of original information during feature refinement. The refined representation $\tilde{\mathbf{X}}$ is then processed through a convolution layer that splits

the output into two distinct branches, maintaining consistency with the two-path design of the overall framework. This structure enables the network to separately retain reflection and moiré-specific characteristics while allowing complementary interactions at the fusion stage. Finally, the module produces the restored image $\mathbf{I}_f$, which is optimized against the ground-truth image $\mathbf{I}_{gt}$ using a fidelity-preserving loss function defined as:

$$L_f(\mathbf{I}_f, \mathbf{I}_{gt}) = \sqrt{\|\mathbf{I}_f - \mathbf{I}_{gt}\|^2 + \epsilon^2}, \quad (3)$$

where ϵ is a small stabilizing constant that prevents numerical instability. This loss encourages the network to generate outputs that are both perceptually accurate and numerically consistent with the reference image.

3. EXPERIMENTS

3.1. Implementation Details

Training Details: The model is trained on a single NVIDIA RTX 3090 GPU card. Training is conducted in two stages. First, the RSM is trained for 500,000 steps using the AdamW optimizer [15] with momentum parameters ($\beta_1 = 0.9, \beta_2 = 0.999$) and a weight decay of 0.02 [16]. To gradually reduce the learning rate from an initial value of $2e^{-4}$ to $1e^{-6}$, we adopt the cosine decay strategy [17]. In the second stage, the Reflection-Moiré Complement Network is trained for an additional 500,000 steps under the same configuration, with the RSM weights kept frozen.

Datasets and metrics: We conduct experiments on the proposed SDD dataset and the synthetic S-SDD data. For quantitative evaluation, we use the PSNR, SSIM [18] and LPIPS [19] metrics to measure the performance of various methods. MAE stands for mean absolute value for the difference between input image and real image. For the PSNR and SSIM metrics, higher values represent better results, while the LPIPS and the MAE metrics lower values represent better results.

3.2. Comparison with State-of-the-Art Methods

To comprehensively address the challenges of screen reflection and moiré removal, we evaluate recent state-of-the-art (SOTA) methods closely related to this task. These include single-image reflection removal [5], the demoiréing method ESDNet [2], and several advanced image restoration methods [6–11]. For fair comparison, we retrain all SOTA methods on our dataset. Table 1 reports the quantitative results on the SDD test set. Our proposed model surpasses all baselines, achieving the highest PSNR and the lowest LPIPS and MAE scores. We further explore combined pipelines that sequentially apply a demoiréing method and a single-image reflection removal method, with results shown in Table 2.



Fig. 2. Visual comparisons between our method and other SOTA methods. (Zoom in to see the details)

Table 1. The quantitative comparison results with the SOTA methods.

Methods	Models trained on the SDD				Models trained on the synthetic data S-SSD				Models complexity	
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MAE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MAE $\downarrow$	Params (M)	FLOPs (G)
Zhu et al. [5]	22.4631	0.8450	0.1053	47.6781	18.1053	0.7976	0.1830	78.6090	28.14	20.34
ESDNet [2]	22.7187	0.8324	0.0859	46.9103	18.0824	0.7651	0.1853	78.4492	10.62	26.89
Restormer [7]	24.1119	0.8715	0.0833	37.9734	17.9303	0.7945	0.1852	83.8282	26.12	87.70
Uformer [6]	24.4306	0.8853	0.0661	37.9136	17.2224	0.7912	0.1854	91.0971	50.88	89.46
FocalNet [11]	24.0751	0.8840	0.0719	39.7405	18.2202	0.8066	0.1715	77.4092	3.74	30.63
NAFNet [10]	22.4242	0.8492	0.1023	48.1441	18.1695	0.7956	0.1846	78.2604	17.11	14.87
CODE [9]	24.8333	0.8875	0.0645	37.0155	17.6170	0.7874	0.1912	87.0436	12.18	179.52
GRL [8]	23.2574	0.8704	0.0779	42.7340	17.6690	0.7787	0.1976	80.1699	20.20	77.63
Ours	25.1026	0.8867	0.0639	34.7921	18.0259	0.7909	0.1896	79.9395	25.81	116.25

Table 2. Combination of demoiréing and SIRR

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MAE $\downarrow$
Demoiré + SIRR	20.2882	0.8142	0.0983	63.4731
SIRR + Demoiré	20.6237	0.8226	0.0957	64.2574
Ours	25.1026	0.8867	0.0639	34.7921

These combinations yield suboptimal performance, likely because artifacts introduced by the first method adversely affect the subsequent stage. To qualitatively assess model performance, Figure 2 presents visual comparisons on SDD. Existing SOTA restoration methods tend to over-suppress highlight regions and fail to fully recover background details, whereas our model effectively removes both reflections and moiré patterns while preserving fine structures. Finally, we investigate the use of synthetic training data by retraining SOTA models on synthesized S-SDD and evaluating them on the SDD test set. The results, summarized in Table 1, show clear performance degradation, highlighting the inability of synthetic data to fully capture real-world degradations.

3.3. Ablation Study

We conduct ablation studies to assess the significance of the RSM, RCM, and the proposed loss functions in our framework. Table 3 presents the quantitative results. First, we remove the RSM and directly input the degraded image into the network. The performance degradation confirms that RSM plays a crucial role in disentangling reflection and transmission layers, which significantly improves image quality. Next, we replace RCM with a spatial fusion block, leading to lower PSNR and higher perceptual distortion, which high-

Table 3. Ablation experiments for the proposed model

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MAE $\downarrow$
Ours w/o RSM	24.5964	0.8784	0.0774	36.3603
Ours w/o RCM	24.9568	0.8835	0.0680	35.1091
Ours w/o L_t	24.0821	0.8612	0.0819	37.8015
Ours w/o L_{tr}	24.1016	0.8605	0.0834	37.0248
Ours	25.1026	0.8867	0.0639	34.7921

lights the efficacy of RCM in jointly handling screen reflections and moiré artifacts. We also evaluate the impact of the loss functions. Removing the transmission loss L_t results in poorer structural fidelity, as the predicted transmission deviates from the ground truth. Eliminating the reconstruction loss L_{tr} weakens the disentanglement constraint, leading to inconsistent layer separation. Collectively, these results validate that both the architectural components (RSM, RCM) and the carefully designed loss functions are indispensable for achieving optimal performance.

4. CONCLUSION

We introduced the Screen Degradation Dataset (SDD), the first real-world benchmark for jointly addressing screen reflections and moiré patterns, and proposed the Reflection-Moiré Complement Network (RMCNet) to effectively handle both degradations. By integrating the Reflection Separation Module (RSM) and the Reflection-Moiré Complement Module (RCM), our method achieves superior restoration compared to SOTA approaches. Experiments on both real and synthetic data confirm the difficulty of real-world degradations and the effectiveness of our framework.

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